FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 16-09-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C, D

Sol. Potential due to non conducting circular disc at a distance of x from its centre is

$$\frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+x^2}-x\right)$$

So, potential at a distance of $\sqrt{3}R$ due to the given portion is

$$V_P=\frac{\sigma}{4\varepsilon_0}\left(\sqrt{R^2+3R^2}-\sqrt{3}R\right)+\frac{\sigma}{4\varepsilon_0}\left(\sqrt{4R^2+3R^2}-\sqrt{3}R\right)=\frac{\sigma R}{4\varepsilon_0}\left[2+\sqrt{7}-2\sqrt{3}\right]$$

Potential at the centre due to the given portion is

$$V_C=\left(\frac{\sigma\cdot 2R}{2\varepsilon_0}\right)\left(\frac{45^\circ}{360^\circ}\right)\times 4+\left(\frac{\sigma\cdot R}{2\varepsilon_0}\right)\left(\frac{45^\circ}{360^\circ}\right)\times 4=\frac{3}{4}\frac{\sigma R}{\varepsilon_0}$$

Work done by external agent = $\frac{3}{4}\frac{\sigma Rq}{\varepsilon_0}$

2. A, B, D

Sol. $$I_x=\frac{m(2\ell)^2}{12}+\frac{m\left(\frac{\ell}{4}\right)^2}{12}=\frac{65}{192}m\ell^2$$

$$I_y=\frac{m(2\ell)^2}{12}+m\left(\frac{\ell}{2}\right)^2+\frac{m\left(\frac{\ell}{4}\right)^2}{12}+m\left(\frac{\ell}{2}\right)^2=\frac{161}{192}m\ell^2$$

$$I_z=\frac{m(2\ell)^2}{6}+m\left(\frac{\ell}{2}\right)^2+m\left(\frac{\ell}{2}\right)^2=\frac{7}{6}m\ell^2$$

3. A, B, C

Sol. $V_{IP} = \left(\frac{f}{f-u}\right)^2 \times \frac{36}{\mu} = \left(\frac{-22.5}{-22.5+90}\right)^2 \times \left(\frac{36}{4/3}\right)$ = 3 cm/s

4. B, C, D

Sol. Let the acceleration of block is a downward and the tension in the string connected with the cylinder is T.

For the block

$64mg - 16T = 64\,ma$ …(i)

For the cylinder

$T + f = 8ma$ …(ii)

Torque about contact point

$T \times 2R = \frac{3}{2}mR^2\alpha$

$R\alpha = 8a$ …(iii)

$T = 6ma$ …(iv)

Solving (i), (ii), (iii) and (iv)

$T = 48N$, $a = 4\ m/s^2$ and $f = 16\ N$

5. A, B, C, D

Sol. $\Delta X_{max} = 2.5\,mm$

$n_{max} = \frac{2.5\times10^{-3}}{5000\times10^{-10}} = 5000$

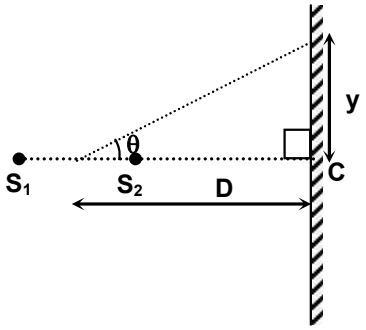
For 3000th order bright fringe, n = 3000

$d\cos\theta = (3000\,\lambda)$

$\cos\theta = \frac{3000\lambda}{5000\lambda} = \frac{3}{5}$

$\frac{D}{\sqrt{y^2+D^2}} = \frac{3}{5}$

Solving $y = \left(\frac{8}{3}\right)m$



For 5th bright fringe from 'C' n = 4996

$5000\,\lambda\cos\theta = 4996\,\lambda$

$\sin\theta = 0.04$

$\tan\theta = y/D = 0.04$

$y = 8\times10^{-2}\ m$

For 2500th order bright fringe, n = 2500

$\cos\theta = 1/2$

$\frac{D}{\sqrt{y^2+D^2}} = \frac{1}{2}$

$4D^2 = y^2 + D^2$

$y = \sqrt{3}D$

$y = 2\sqrt{3}\ m$

6. A, C

Sol. ${}_{92}U^{238} \rightarrow {}_{90}Th^{234} + {}_{2}He^{4}$

$\frac{P^2}{2m_\alpha} + \frac{P^2}{2m_{Th}} = Q$

$$P = \left(\frac{2m_\alpha m_{Th} Q}{m_\alpha + m_{Th}}\right)^{1/2}$$

$$\frac{K.E_\alpha}{K.E_{Th}} = \frac{m_{Th}}{m_\alpha} = \frac{234}{4} = \frac{117}{2}$$

7. A, C

Sol. Torque about P_1P_2

$$i\ell^2 B \sin 60^\circ = \frac{2}{3} m\ell^2 \alpha$$

$$\alpha = \frac{3\sqrt{3}}{4}\frac{Bi}{m}$$

$$\frac{1}{2} I\omega^2 = U_i - U_f = -Bi\ell^2 \cos 60^\circ - (-Bi\ell^2 \cos 0^\circ)$$

$$\frac{1}{2} I\omega^2 = \frac{Bi\ell^2}{2}$$

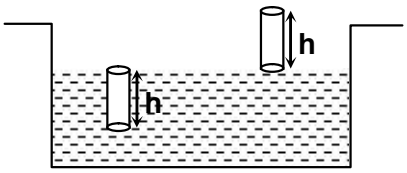
$$\Rightarrow \omega = \sqrt{\frac{Bi\ell^2}{I}} = \sqrt{\frac{Bi\ell^2 \times 3}{2m\ell^2}} = \sqrt{\frac{3Bi}{2m}}$$

8. A, C, D

Sol. $$(Ah)(\rho_\ell)\left[g\frac{h}{2}\right] = (Ah)(\rho_S g)\left(\frac{h}{2} + \frac{h}{2}\right) + \frac{1}{2}(Ah)(\rho_S)v^2$$

$$\frac{\rho_\ell gh}{2} = \rho_s gh + \frac{1}{2}\rho_s v^2$$

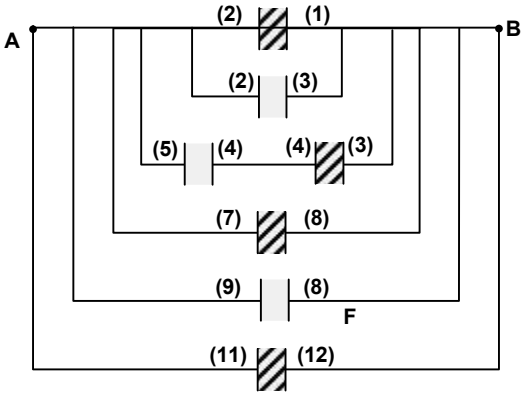
$$v = \sqrt{\frac{(\rho_\ell - 2\rho_s)gh}{\rho_s}}$$



9. D

10. A

Sol. Apply Kirchoff's Law on the equivalent circuit shown in the diagram.



11. C

12. B

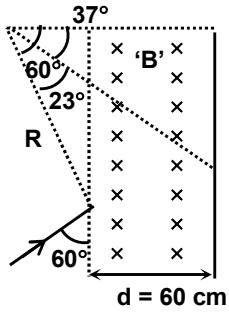
Sol. (for Q.11 – Q.12): Particle collides with the wall at its maximum height hence, the time after which the ball collides with the wall is $t = \frac{20\sqrt{2}\sin 45^\circ}{10} = 2\,\text{sec}$

SECTION – C

13. 00000.25

Sol. $d = R\cos 37° - R\cos 60°$

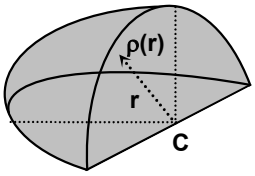
$R = 2$ m

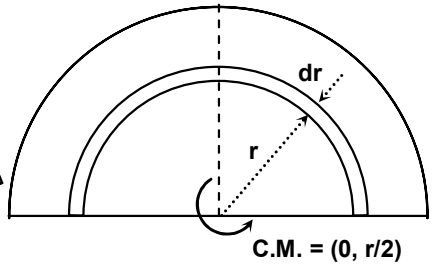


14. 00003.50

Sol. $$y_{cm} = \frac{\int_0^R \left[\rho_0\left(1-\frac{r^2}{R^2}\right)(2\pi r^2 dr)\right]\frac{r}{2}}{\int_0^R \rho_0\left(1-\frac{r^2}{R^2}\right)(2\pi r^2)dr} = \frac{5}{16}R$$

Position of CM, from C $= \left(\frac{5\sqrt{2}}{16}R\right)$





15. 00054.46

Sol. Change of resistance by 1 Ω = change of deflection by 4 + 6 = 10

Change in deflection by 1 division = change of resistance by 1/10 Ω

Change in deflection by 4 division = change of resistance by 4/10 Ω = 0.4 Ω

Here for null point R = 545 – 0.4 = 544.6 Ω

So, unknown resistance = $\frac{544.6\times 10}{100} = 54.46\ \Omega$

16. 00002.80

Sol. $$\vec{B} = \frac{\mu_0}{4\pi}\frac{q(\vec{v}\times\vec{r})}{r^3}$$

$$\vec{B} = \mu_0\varepsilon_0(\vec{v}\times\vec{E}) = \mu_0\varepsilon_0(-\hat{i}+\hat{j}-\hat{k})\times(\hat{i}+\hat{j}+\hat{k})$$

$$\vec{B} = 2(\hat{i}-\hat{k})\mu_0\varepsilon_0$$

$$B = 2\sqrt{2}\mu_0\varepsilon_0$$

$$K = 2\sqrt{2} = 2.80$$

17. 00020.00

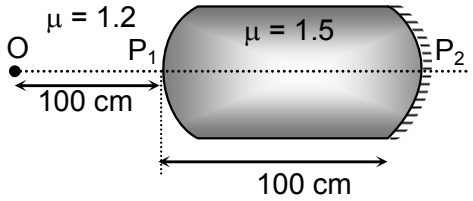
Sol. Refraction from P_1

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2-\mu_1}{R}$$

$$\frac{1.5}{v_1} - \frac{1.2}{-100} = \frac{1.5-1.2}{+20}$$

$\Rightarrow v_1 = 500$ cm

Reflection from P_2



$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

$$\frac{1}{-200}=\frac{1}{v_2}+\frac{1}{400}, \quad v_2=\frac{-400}{3}\text{cm}$$

Again refraction from P_1

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}, \quad \frac{1.2}{v_3}-\frac{1.5}{100/3}=\frac{1.2-1.5}{-20}$$

$\Rightarrow v_3 = +20$ cm (left to P_1)

18. 00003.52

Sol. 8S = 10 V

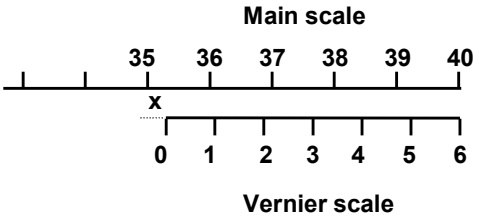
V = 0.8 mm

6V = 0.8 × 6 = 4.8 mm

40S = 35S + x + 6V

x = 0.2 mm

Measured length = 3.52 cm



Chemistry

PART – II

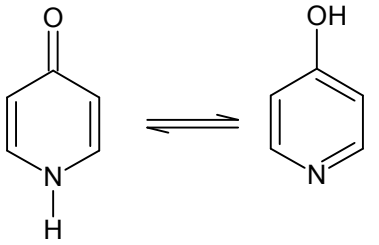
SECTION – A

19. B, C, D

Sol. Process is endothermic in forward direction.

20. A, C, D

Sol. $R_2CHCH=NH \rightleftharpoons R_2C=CH-NH_2$



$CH_3CH_2NO \rightleftharpoons CH_3CH=NOH$

21. A, B, C

Sol. Absence of POS and COS.

22. A, B, C, D

23. A, B, C, D

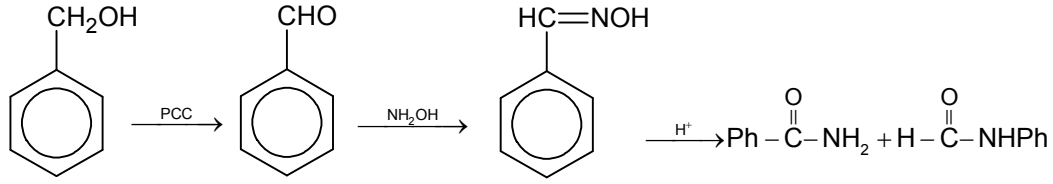
Sol. Facts

24. A, B, C, D

Sol. Facts.

25. A, B, C, D

Sol.



$$Ph-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[H_2O]{Br_2/Alc.\ KOH} PhNH_2 \xrightarrow[Alc.\ KOH]{CHCl_3} PhNC$$

$$H-\overset{O}{\overset{\|}{C}}-NHPh \xrightarrow{H_3O^+} PhNH_2 + HCOOH$$

26. A, B, C

Sol. Facts

27. A

28. D

Sol. (for Q.27 – Q.28):

$$Pb^{2+} + KI \longrightarrow \underset{\text{Yellow}}{PbI_2\downarrow} \xrightarrow{KI} K_2PbI_4$$

$Ag^{+} + KI \longrightarrow \underset{\text{Yellow}}{AgI\downarrow}$

$Cu^{2+} + KI \longrightarrow \underset{\text{White}}{Cu_2I_2\downarrow}$

$Cu^{2+} + K_4[Fe(CN)_6] \longrightarrow \underset{\text{Brown}}{Cu_2[Fe(CN)_6]\downarrow}$

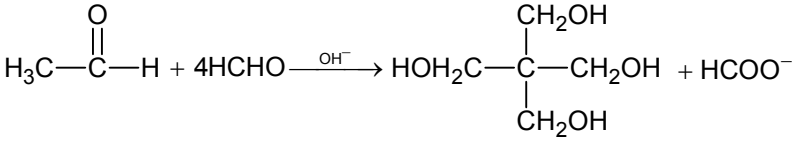
$Cu^{2+} + KCN \longrightarrow \underset{\text{Yellow}}{Cu(CN)_2\downarrow} \longrightarrow \underset{\text{White}}{CuCN\downarrow} + (CN)_2$

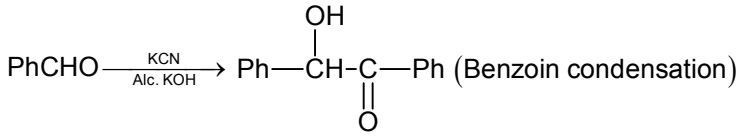
$Bi^{3+} + KI \longrightarrow \underset{\text{Black}}{BiI_3\downarrow} \xrightarrow{KI} \underset{\text{Orange solution}}{KBiI_4}$

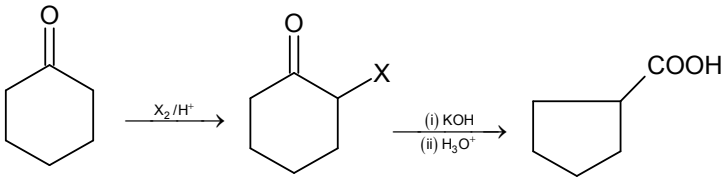
29. D

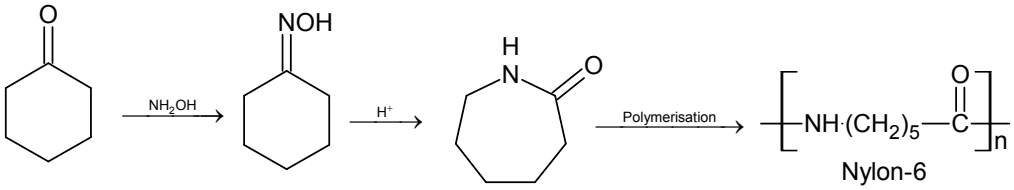
30. A

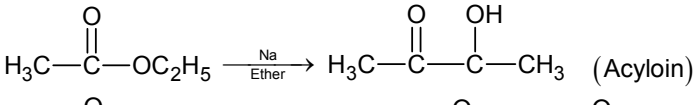
Sol. (for Q.29 – Q.30):

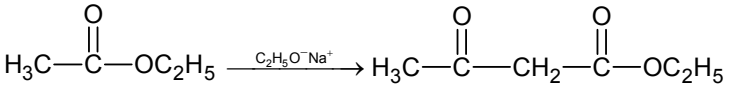












(Claisen condensation)

SECTION – C

31. 00030.00

Sol. Total possible are 30.00.

32. 00004.20

Sol. $i = 1 + (n - 1)\alpha$ where n = 5

$= 1 + [5 - 1]0.8 = 4.2$

33. 00003.75

Sol. Removed 1 Zn^{2+} from tetrahedral void.

$\therefore$ Remaining occupied tetrahedral void = 3

$$\text{Fraction} = \frac{3}{8} = 0.375$$

$$\therefore 10X = 3.75$$

34. 00024.44

Sol. $W_{CO_2} = 1.44$ gm

$$\therefore \text{Moles of } CO_2 = \frac{1.44}{44} = \text{moles of } CO_3^{2-}$$

$$\therefore W_{CO_3^{2-}} = \frac{1.44}{44} \times 60 = 1.9636$$

Let two metal CO_3^{2-} are ACO_3 and BCO_3.

If A is 14.2%

$$\% \text{ of B} = \frac{3.2 - [1.9636 + 0.4544]}{3.2} \times 100$$

$$= 24.437 \approx 24.44\%$$

35. 00011.51

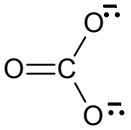
Sol. $$\left[\bar{O}H\right] = \frac{(80 \times 0.1) - (75 \times 0.1)}{155} = 0.0032258$$

$= 3.23 \times 10^{-3}$

pOH = 2.49

$\therefore$ pH = 11.51

36. 00001.33

Sol. 

1.33

Mathematics

PART – III

SECTION – A

37. B, C

Sol. $Z = \sum_{r=1}^{100}(2r+3)(i)^r = 2\sum_{r=1}^{100} r\,i^r + 3\sum_{r=1}^{100} i^r$

$Z = 100(1 - i)$

38. A, B

Sol. $x^n - a^n$ is divisible by $x - a$ if n is odd

39. A, B, D

Sol. Midpoint of PQ is $\left(\frac{7}{2}, \frac{9}{2}\right)$

$\therefore$ Slope of axis is $\frac{9}{7}$

Hence equation of directrix is $9y + 7x = 0$

Equation of PQ is $11x - 3y - 25 = 0$ and foot of perpendicular from origin on PQ is focus which is $\left(\frac{55}{26}, -\frac{15}{26}\right)$

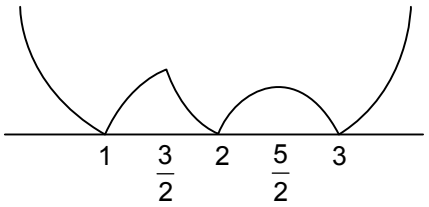
Foot of perpendicular from focus S on the line $y - 2x = 0$ is $\left(\frac{5}{26}, \frac{10}{26}\right)$

$\therefore$ Equation of tangent at vertex is $y - \frac{10}{26} = -\frac{7}{9}\left(x - \frac{5}{26}\right)$

i.e., $182x + 234y - 125 = 0$

40. B, C

Sol. Use graph of f(x)



41. A, B, C

Sol. Let $\hat{x}\cdot\hat{y} = a$, $\hat{y}\cdot\hat{z} = b$, $\hat{x}\cdot\hat{z} = c$

$|\hat{x} + \hat{y} - \hat{z}| = |\vec{p}|$

$\Rightarrow 3 + 2(a - b - c) = 3$

$\Rightarrow a - b - c = 0$ ..... (1)

Also, $\hat{x}\times(\hat{y}\times\hat{z}) = \vec{q}$

$\Rightarrow c\hat{y} - a\hat{z} = \vec{q}$ ..... (2)

And $(\hat{x}\times\hat{y})\times\hat{z} = \vec{r}$

$\Rightarrow c\hat{y} - b\hat{x} = \vec{r}$ ..... (3)

From equation (2), $c - ab = \frac{3}{4}$

$\because \vec{q}\cdot\vec{y} = \frac{3}{4}$

Also, $\vec{p}\cdot\vec{x} = 1 + a - c \Rightarrow a - c = -1$

Using equation (1), b = –1 and $c + a = \frac{3}{4} \Rightarrow a = -\frac{1}{8}, c = \frac{7}{8}$

$\therefore \vec{r} = \frac{7}{8}\hat{y} + \hat{x}$; $\vec{q} = \frac{7}{8}\hat{y} + \frac{\hat{z}}{8}$

$|\vec{q}| = \left|\frac{7\hat{y} + \hat{z}}{8}\right| = \frac{\sqrt{49 + 1 + 14b}}{8} = \frac{3}{4}$

42. A, B

Sol. $P(x) = x^4 + px^3 + qx^2 + rx - 48$

Let $Q(x) = x^3 - 2ax^2 + bx - 6$

$R(x) = x^3 + ax^2 - (3b + 1)x + 48$

Q(x) and R(x) must have 2 common roots (β, γ)

Which satisfy $3ax^2 - (4b + 1)x + 54 = 0$

$\Rightarrow \beta\gamma = \frac{18}{a}$; $a \neq 0$

$\alpha\beta\gamma = 6$, $\beta\gamma\delta = -48$ and $\alpha = 1$

Also, $\alpha\beta\gamma\delta = -48 \Rightarrow \delta = -8$

$\therefore \beta\gamma = 6$

Hence, a = 3

Also, $\alpha + \beta + \gamma = 2a \Rightarrow \beta + \gamma = 5$

$\therefore \beta = 2, \gamma = 3$

$\therefore P(x) = (x - 1)(x - 2)(x - 3)(x + 8)$

a = 3, b = 11

43. A, C, D

Sol. $L_1 = \lim\limits_{x\to 0} \frac{3ax + (a-8)\sin x}{\left(\sin^{-1} x\right)^3} = \frac{(4a-8)x - \frac{1}{6}(a-8)^3 x^3 + ...}{\left(\sin^{-1} x\right)^3}$. If L_1 is finite then a = 2

$L_2 = \lim\limits_{x\to\infty} \frac{8 - \left(\frac{\text{cosec}^{-1} x}{x}\right)^5}{\frac{4}{\pi}\sec^{-1}(kx - 1) + \frac{k^3}{\pi}\left(\frac{\tan^{-1}\frac{\pi}{x^5}}{\frac{1}{x^5}}\right) + \frac{5k^2}{2}}$. If k = 0 then $L_2 = \frac{8-0}{\frac{4}{\pi}\pi} = 2$

If $k \neq 0$ then $L_2 = \frac{8-0}{\frac{4}{\pi}\frac{\pi}{2} + k^3 + \frac{5k^2}{2}} = 2$

$\Rightarrow 2k^3 + 5k^2 - 4 = 0 \Rightarrow k = -2$

44. A, C, D

Sol. $4\alpha - \beta + 3\gamma = 4$ ..... (1)

$\alpha + 3\beta - 4\gamma = 2$ ..... (2)

$7\alpha + 8\beta - 9\gamma = 10$ ..... (3)

Given system of equations has infinite solution where $\alpha = \frac{14 - 5\gamma}{13}$, $\beta = \frac{4 + 19\gamma}{13}$

Moreover equation of plane is 3x + 4y + 5z + 44 = 0

$\because$ P(α, β, γ) lies on the plane $\therefore \alpha = 3, \beta = -7, \gamma = -5$

45. A

46. D

Sol. (for Q.45 – Q.46):

$\text{Re}((1+i)Z^2) > 0$

$\Rightarrow x^2 - y^2 - 2xy > 0$

$\Rightarrow -(\sqrt{2}+1)x < y < (\sqrt{2}-1)x \; ; \; x \geq 0$

$(\sqrt{2}-1)x < y < -(\sqrt{2}+1)x \; ; \; x < 0$

$\text{Im}((1+i)Z^2) > 0 \; ; \; y < 0$

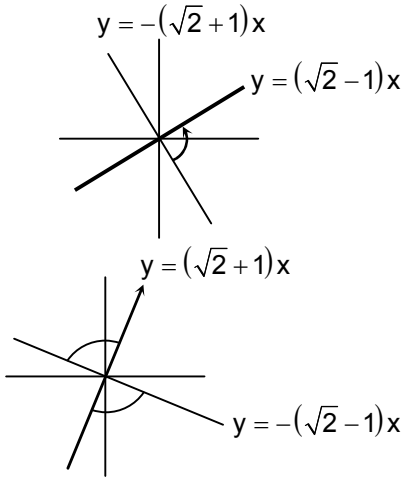
$\Rightarrow x^2 - y^2 + 2xy > 0$

$\Rightarrow -(\sqrt{2}-1)x < y < (\sqrt{2}+1)x$

$\text{Im}(1+i)Z^2 < 0 \; ; \; y \geq 0$

$\Rightarrow x^2 - y^2 + 2xy < 0$

$\Rightarrow y < -(\sqrt{2}-1)x$ or $y > (\sqrt{2}+1)x$



47. B

48. C

Sol. (for Q.47 – Q.48):

$f(x) = \begin{cases} -2 & ; \; -1 \leq x < -\frac{1}{2} \\ -1 & ; \; -\frac{1}{2} \leq x < 0 \\ 0 & ; \; 0 \leq x < \frac{1}{2} \\ 1 & ; \; \frac{1}{2} \leq x < 1 \\ 2x & ; \; 1 \leq x \leq 2 \end{cases}$

$f(f(x)) = \begin{cases} -2 & ; \; -\frac{1}{2} \leq x < 0 \\ 0 & ; \; 0 \leq x < \frac{1}{2} \\ 2 & ; \; \frac{1}{2} \leq x < 1 \\ 4 & ; \; x = 1 \end{cases}$

$f(2g(x)) = \begin{cases} -2 & ; \; x = \frac{3\pi}{2} \\ 2\,\text{cosec}\,x & ; \; x \in \left[\frac{\pi}{6}, \frac{5\pi}{6}\right] \end{cases}$

$f(g(x)) = \begin{cases} -2 & ; \; x \in \left[\frac{7\pi}{6}, \frac{11\pi}{6}\right] - \left\{\frac{3\pi}{2}\right\} \\ -1 & ; \; x = \frac{3\pi}{2} \\ 1 & ; \; x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right) \\ \text{cosec}\,x & ; \; x \in \left(0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, \pi\right) \end{cases}$

$g(f(x)) = \begin{cases} \frac{\text{cosec}\,1}{2} & ; \; \frac{1}{2} \leq x < 1 \\ \frac{\text{cosec}\,2x}{2} & ; \; 1 \leq x \leq 2 - \left\{\frac{\pi}{2}\right\} \end{cases}$

SECTION – C

49. 00006.00

Sol. $\sin^2\frac{\pi}{n}+\sin^2\frac{2\pi}{n}=\sin^2\frac{3\pi}{n}$

$\Rightarrow \sin^2\frac{2\pi}{n}=\sin\frac{4\pi}{n}\cdot\sin\frac{2\pi}{n}$

$\Rightarrow \sin^2\frac{2\pi}{n}\left(2\cos\frac{2\pi}{n}-1\right)=0$

$\Rightarrow \cos\frac{2\pi}{n}=\frac{1}{2} \Rightarrow n = 6$

50. 00005.00

Sol. Required probability = $\dfrac{\frac{4}{5}\times\frac{1}{10}}{\frac{4}{5}\times\frac{1}{10}+\frac{1}{5}\times 1}=\frac{2}{7}$

51. 00004.50

Sol. Equation reduces to $\sin\frac{4}{3}x+\cos\frac{16}{9}x=2$

$\Rightarrow \sin\frac{4}{3}x=1$ and $\cos\frac{16}{9}x=1$

$\Rightarrow x=(4n+1)\frac{3\pi}{8}$ and $x=\frac{9m\pi}{8}$

$\Rightarrow m=\frac{4n+1}{3} \Rightarrow n = 3k + 2;\ k \in I$

$\therefore\ x=(4k+3)\frac{9\pi}{8}$; k has 4 values

52. 00001.50

Sol. $f(x) = \cos^{-1} x \qquad -\frac{1}{2}\le x<\frac{1}{2}$

$\sin^{-1}(1-x) \qquad \frac{1}{2}\le x<\frac{3}{2}$

$\cos^{-1}(x-2) \qquad \frac{3}{2}\le x<\frac{5}{2}$

$\sin^{-1}(3-x) \qquad \frac{5}{2}\le x<\frac{7}{2}$

f(x) is periodic with period 2

$$\int_{-\frac{1}{2}}^{\frac{23}{2}} f(x)\,dx = 6\left[\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos^{-1}x\,dx+\int_{\frac{1}{2}}^{\frac{3}{2}} \sin^{-1}(1-x)\,dx\right] = 3\pi$$

53. 00009.00

Sol. Let the centre be $(2\cos\theta, 3\sin\theta)$

$\frac{x_1+x_2+x_3+x_4}{4}=\frac{2\cos\theta}{2}$ and $\frac{y_1+y_2+y_3+y_4}{4}=\frac{3\sin\theta}{2}$

If (h, k) is the centroid of Δ then $x_4 = 4\cos\theta - 3h$, $y_4 = 6\sin\theta - 3k$

Hence, $\left(h-\frac{4}{3}\cos\theta\right)\left(k-2\sin\theta\right)=\frac{2}{9}$

54. 00003.25

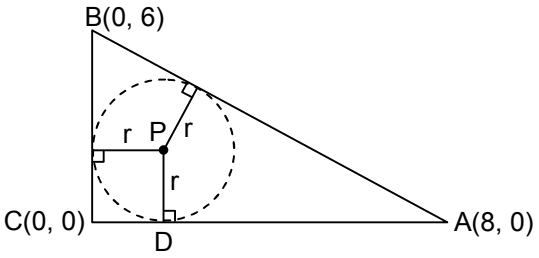
Sol. $\frac{1}{2}(r+a)r=8$

$\Rightarrow r = 2$

AB = b + 2

$\because (b+2)^2 = b^2 + 6^2$

$\Rightarrow b = 8$



Let equation of required circle is

$(x-h)^2 + (y-3)^2 = h^2 + 9$

$\because$ Incircle and the required circle touch internally

$\therefore \left(h-2\right)^2+1=\left|\sqrt{h^2+9}-2\right|^2 \Rightarrow h=\frac{5}{4}$ $\therefore r_1=\sqrt{h^2+9}=\sqrt{\frac{25}{16}+9}=\frac{13}{4}$